

Regression Exercise Questions

1. Please enter the P -values for marker rs1112 from the analyses in the table below.

	Type of Analysis	P -value
I.	Single marker, case-control, genotypic model	<u>Wald test:</u> het 1/2: hom 2/2: <u>LRT:</u>
	Single marker, case-control, allelic model	<u>Wald:</u> <u>LRT:</u>
II.	Single marker, case-control, adjustment for age	
	Single marker, case-control, adjustment for sex	
	Single marker, case-control, adjustment for sex & age	
	Single marker, case-control, adjustment for marker rs1117	<u>Wald:</u> <u>LRT:</u>
III.	Single marker, quantitative trait, adjustment for sex	
IV.	Interaction P -value with covariate sex	
	Interaction P -value with marker rs1117	

2. Please give the odds ratio (OR) and its 95% confidence interval for marker rs1112 in the unadjusted, genotype-based case-control analysis.

$OR_{het(1/2)} = \dots\dots\dots$ 95% CI = $\dots\dots\dots - \dots\dots\dots$
 $OR_{hom(2/2)} = \dots\dots\dots$ 95% CI = $\dots\dots\dots - \dots\dots\dots$

3. In the combined analysis of rs1112 and rs1117 (section II., no interaction), the LRT-based P -values strongly depended on the order of the markers in the regression model (affection ~ rs1117+rs1112: $p_{rs1117}=5.547e-07$ / $p_{rs1112}=1.193e-03$; affection ~ rs1112+rs1117: $p_{rs1112}=5.438e-09$ / $p_{rs1117}=0.21$). Do you have an explanation?